

Physical Chemistry (I)

Lecture 20



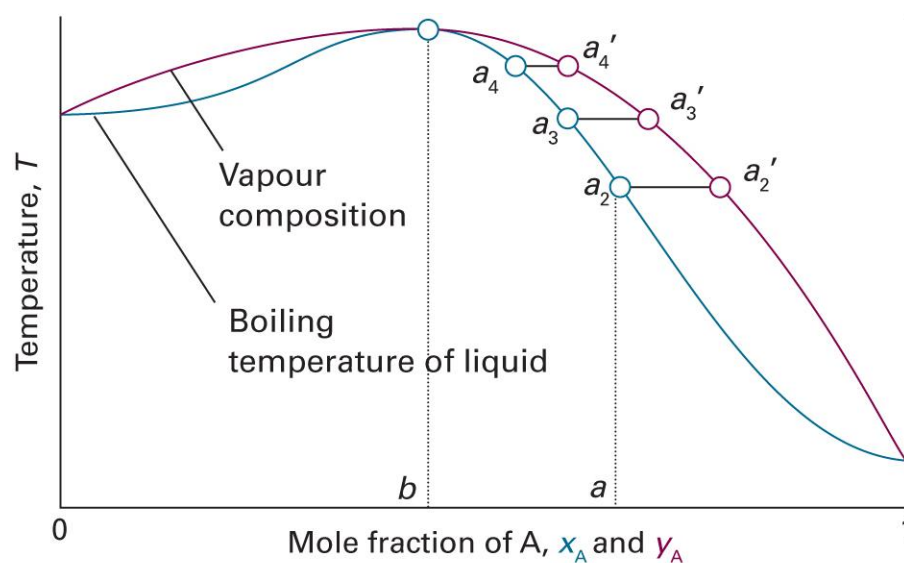
In the Last Class

- Phase diagrams of various systems
 - Real solutions: azeotrope, phase separation
 - Liquid-solid mixtures
- Liquid-solid systems: colligative properties
 - Lowering of vapor pressure
 - Elevation of boiling point, depression of freezing point
 - Solubility
 - Osmotic pressure

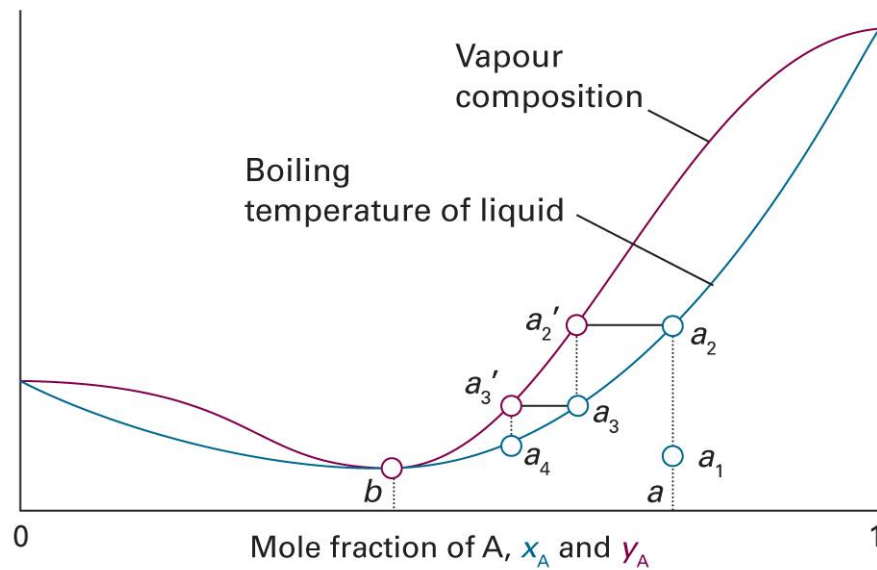


Azeotropes

A-B interactions stronger

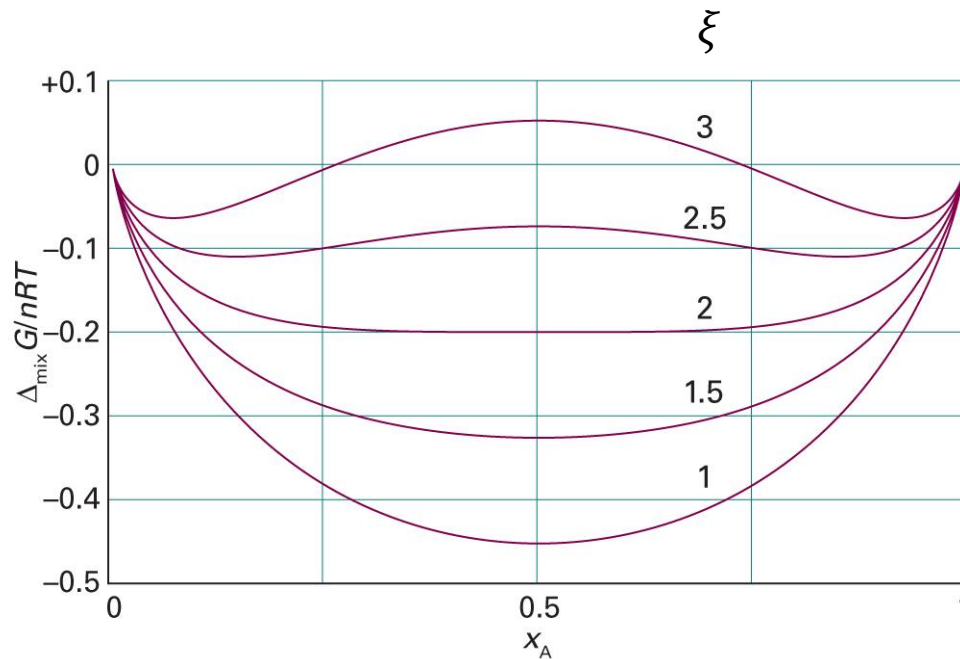


A-B interactions weaker



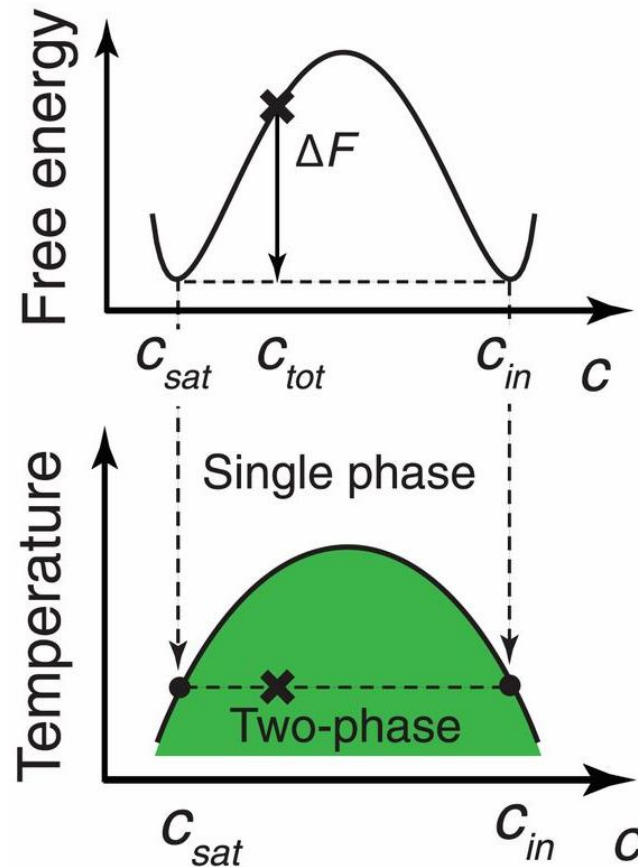
BW Model: Gibbs Energy of Mixing

$$\Delta_{\text{mix}}G = nRT(x_A \ln x_A + x_B \ln x_B + \xi x_A x_B)$$



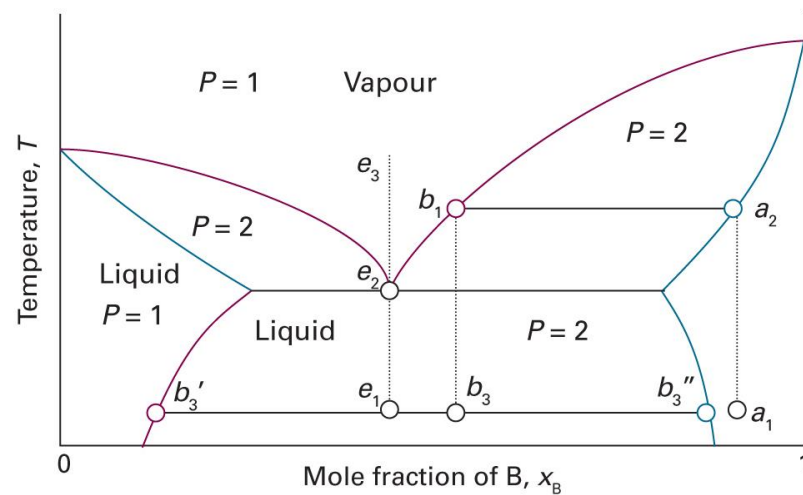
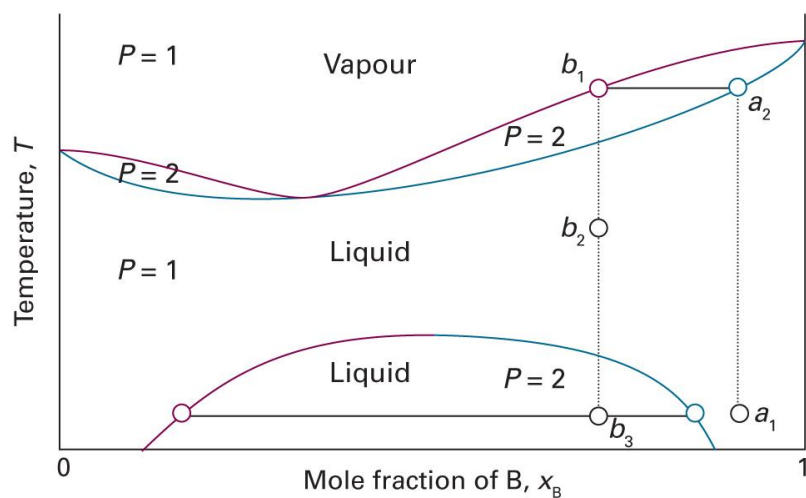
(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figure 5B.5)

Liquid-Liquid Phase Separation



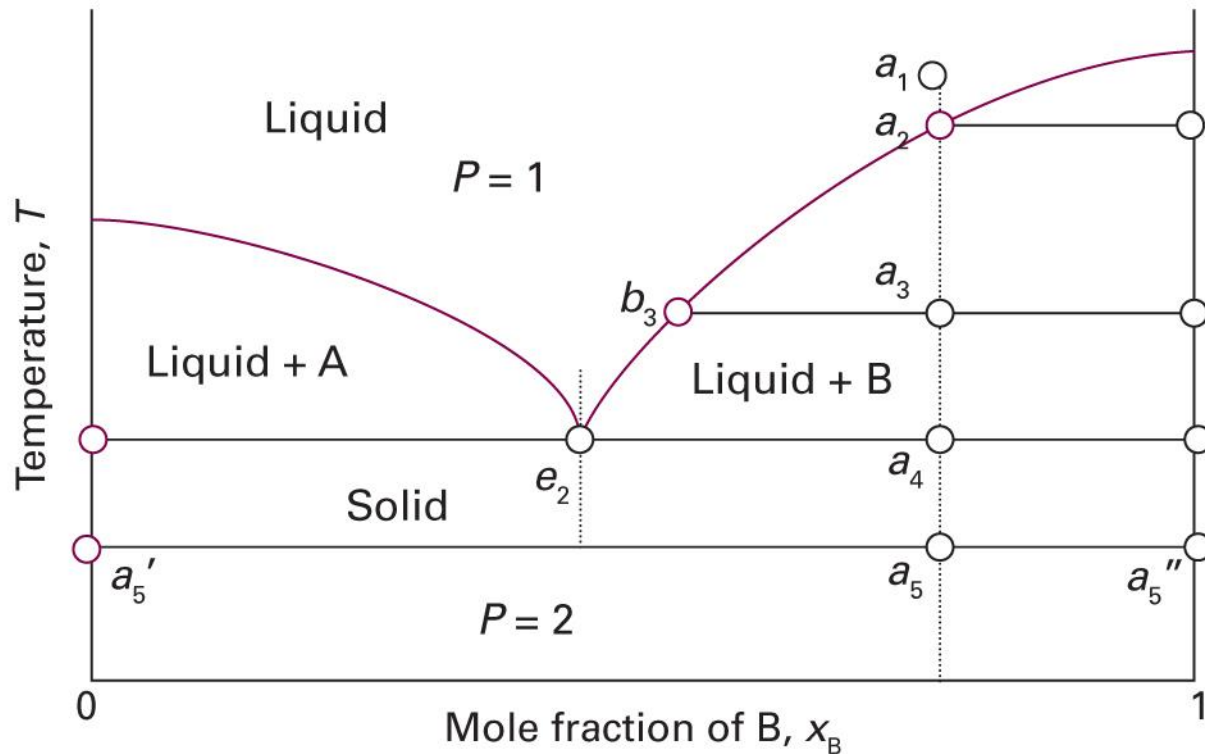
(Image: Shin and Brangwynne, *Science* 357: eaaf4382 (2017), Figure 3)

Various Phase Diagrams



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figures 5C.20 and 5C.21)

Phase Diagrams Involving Solids



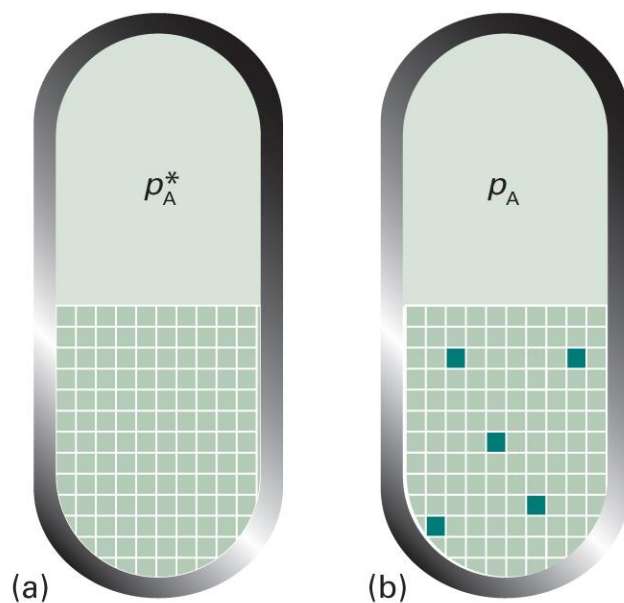
(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figure 5D.1)

Colligative Properties

- Physical properties that depends on the relative number of solute particles present but not their chemical identity
1. Lowering of vapor pressure
 2. Elevation of boiling point
 3. Depression of freezing point
 4. Osmotic pressure

Origin of Colligative Properties

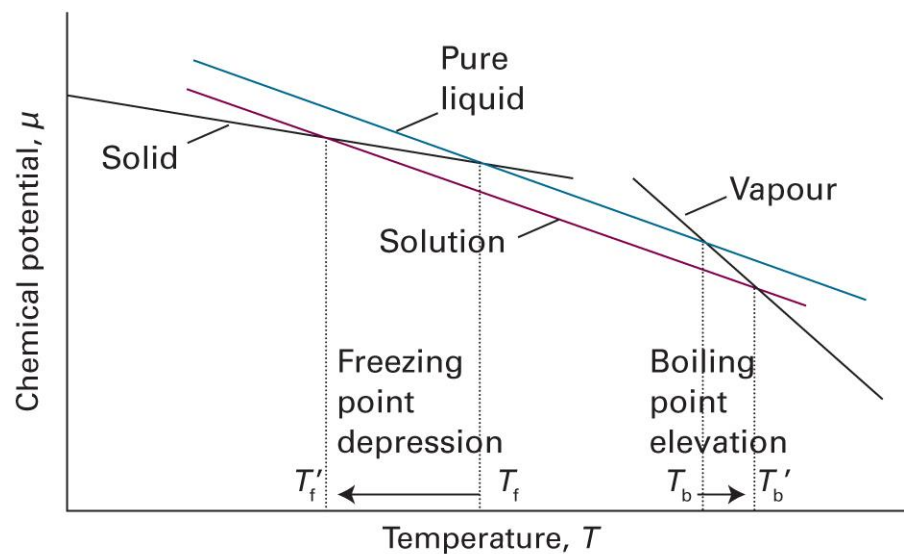
- Due to the **reduction of the chemical potential** of the liquid solvent as a result of the presence of solute



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figure 5B.7)

Changes in Phase Transitions

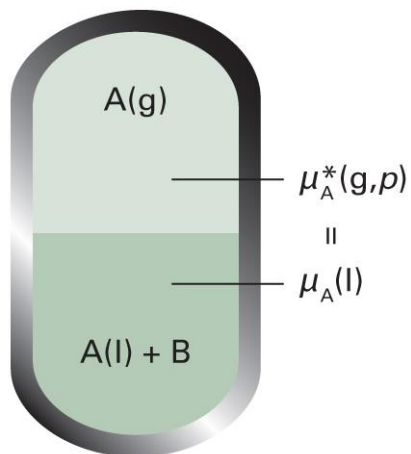
$$\mu_A = \mu_A^* + RT \ln x_A \quad (x_A < 1)$$



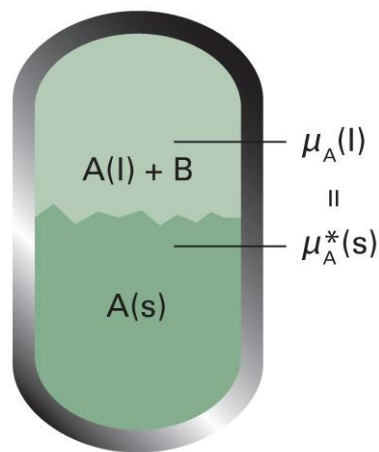
(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figure 5B.6)

Quantitative Treatments

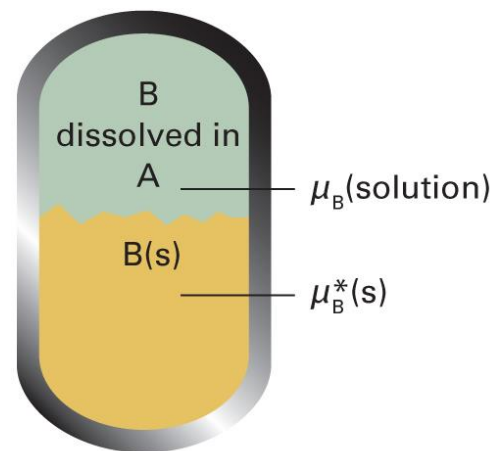
Boiling point elevation



Freezing point depression



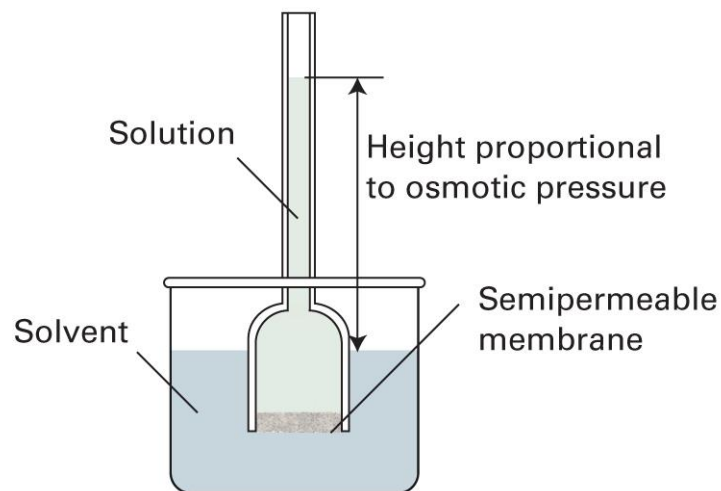
Solubility



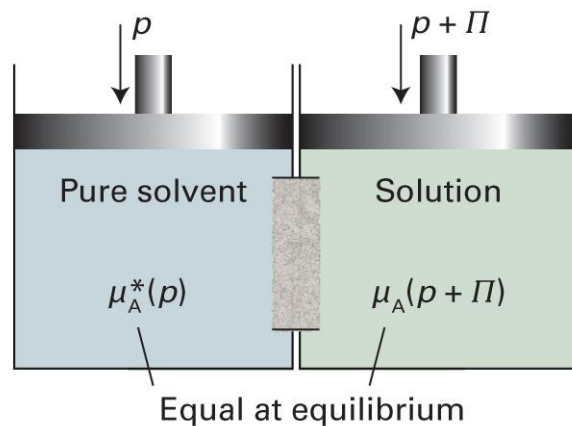
(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figures 5B.8, 5B.9, and 5B.10)

Osmosis

spontaneous passage of a pure solvent into a solution separated from it by a semipermeable membrane



Thermodynamics of Osmosis



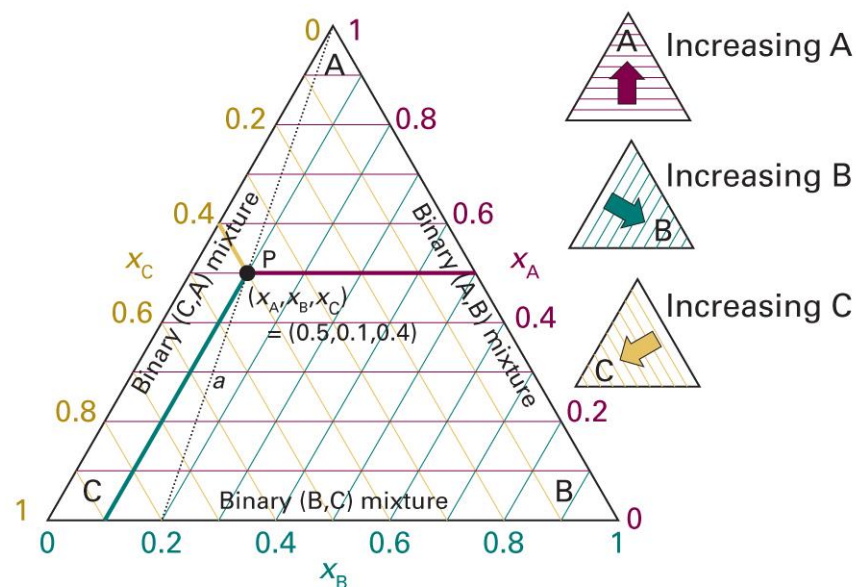
$$\Pi = [J]RT \text{ (van't Hoff equation)}$$

Real solution: osmotic virial expansion

$$\Pi = [J]RT(1 + B[J] + \dots)$$

Ternary Systems: Triangle

$$x_A + x_B + x_C = 1$$

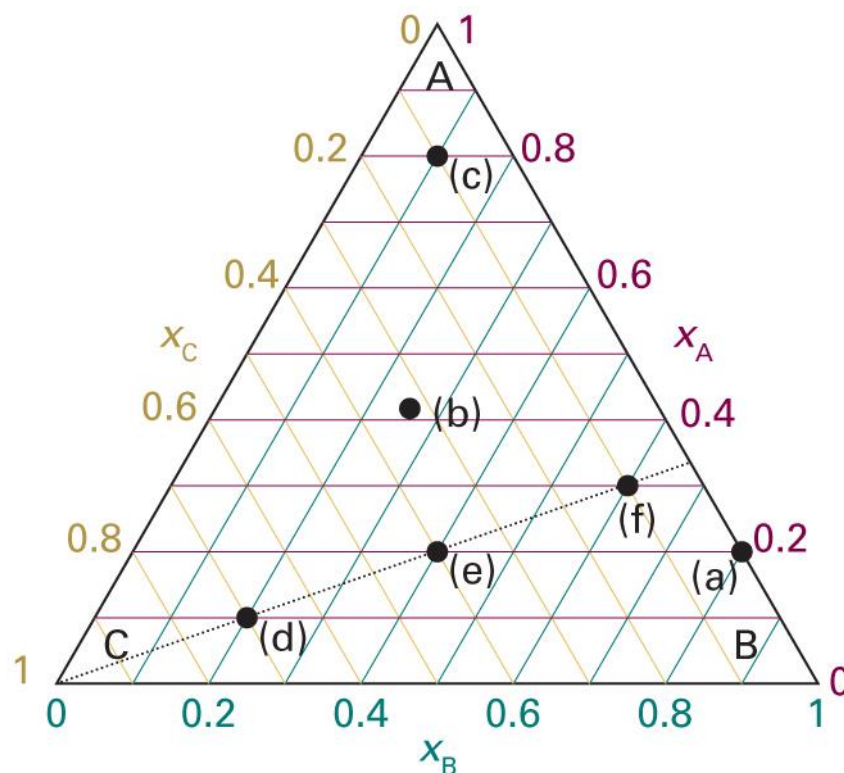


(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figure 5E.1)

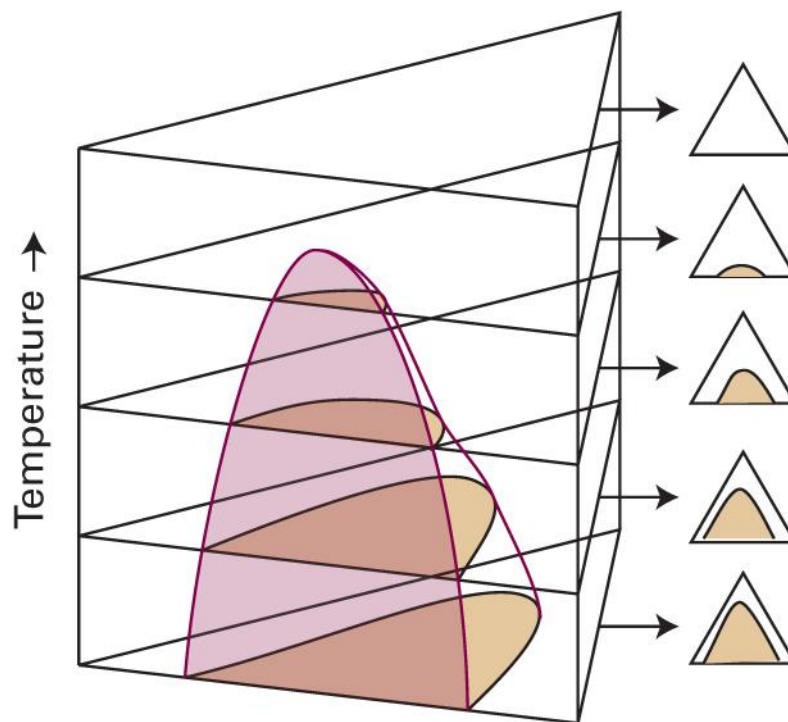
Brief Illustration 5E.1

Indicate the following points:

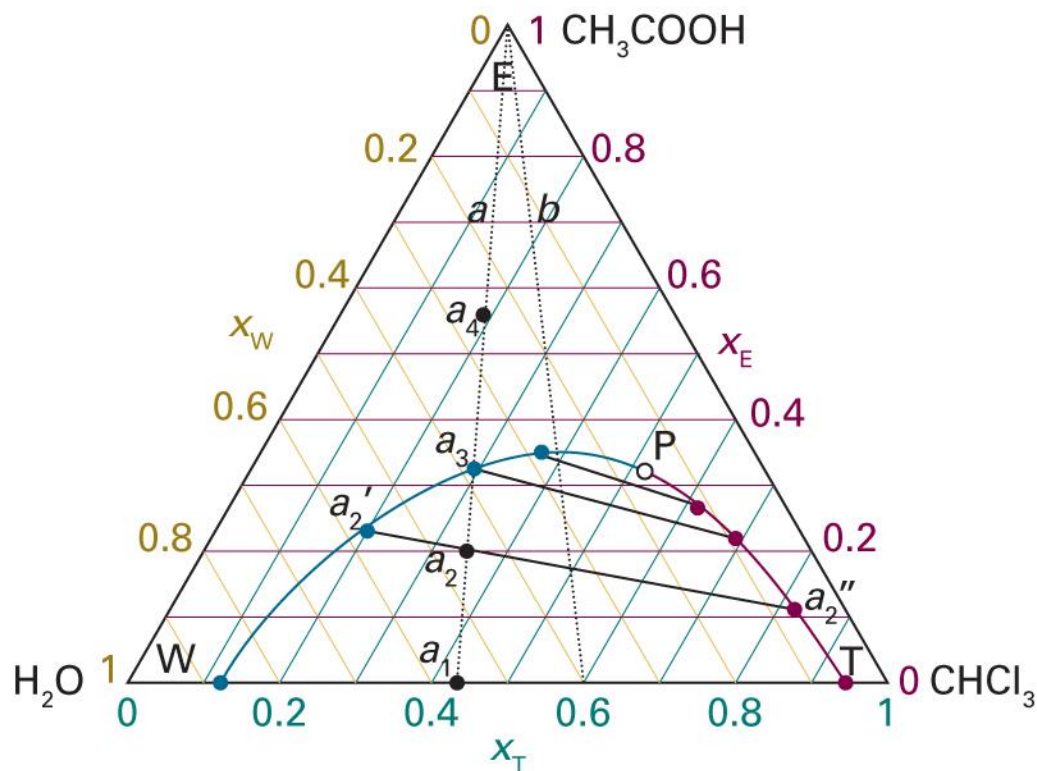
Point	x_A	x_B	x_C
a	0.20	0.80	0
b	0.42	0.26	0.32
c	0.80	0.10	0.10
d	0.10	0.20	0.70
e	0.20	0.40	0.40
f	0.30	0.60	0.10



What If Temperature Changes?

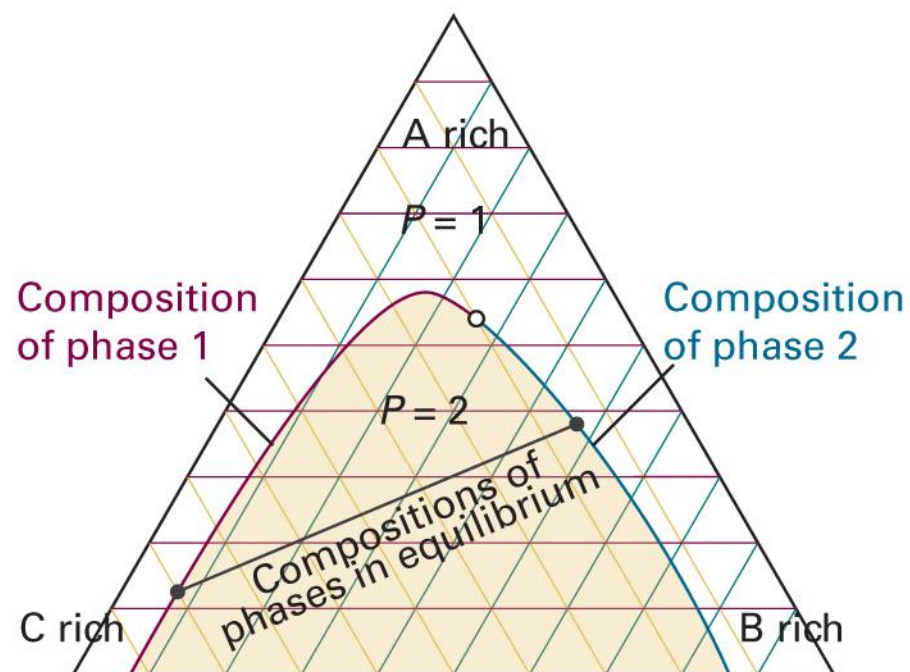


Partially Miscible Liquids

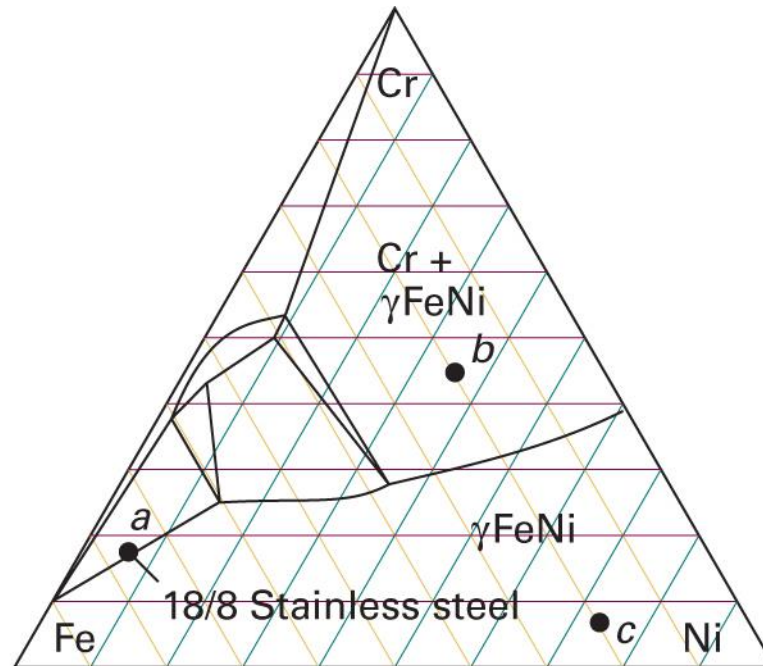


(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figure 5E.4)

Interpretation



Ternary Solids



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figure 5E.6)

Summary

- Phase
- Chemical potential
- Gibbs-Duhem eq.

Multi-component
system

Ternary systems

Gas mixture

Dalton's law

Perfect gas

- Chemical potential
$$\mu = \mu^\ominus + RT \ln(p/p^\ominus)$$
- $\Delta_{\text{mix}}G = -T\Delta_{\text{mix}}S$

Real gas

- Chemical potential
$$\mu = \mu^\ominus + RT \ln(f/p^\ominus)$$
- Fugacity f
- $\Delta_{\text{mix}}G \neq -T\Delta_{\text{mix}}S$

Liquid mixture

Ideal solution

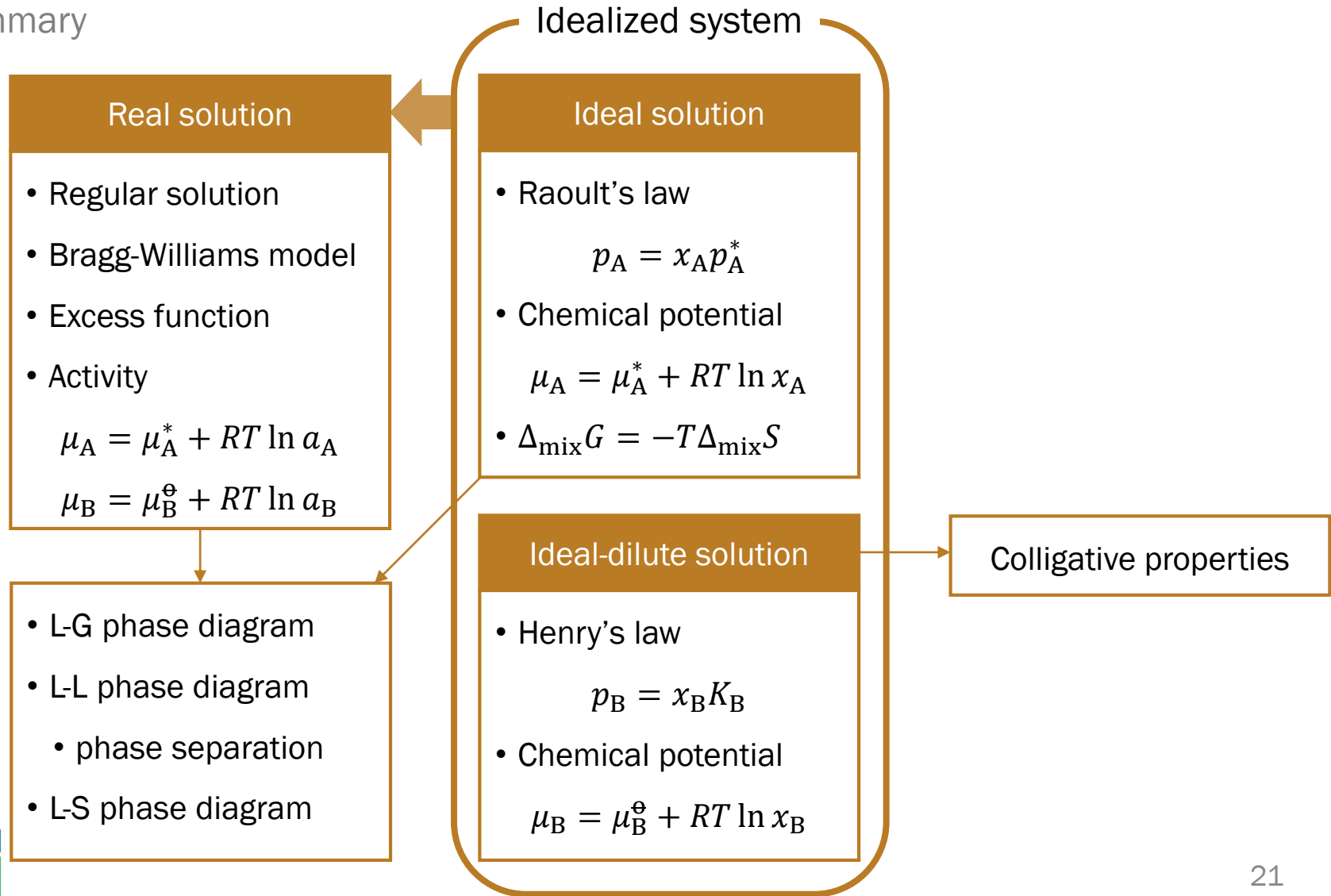
- Raoult's law
$$p_A = x_A p_A^*$$
- Chemical potential
$$\mu_A = \mu_A^* + RT \ln x_A$$
- $\Delta_{\text{mix}}G = -T\Delta_{\text{mix}}S$

Ideal-dilute solution

- Henry's law
$$p_B = x_B K_B$$
- Chemical potential
$$\mu_B = \mu_B^\ominus + RT \ln x_B$$

...

Summary



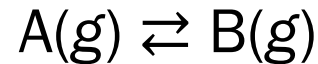
8. Chemical Equilibria

Chapter Overview

- Chemical reactions and equilibria
 - Reaction quotient and equilibrium constant
 - Activities
- Le Chatelier's principle
 - Amount change
 - Volume change
 - Temperature change



Simple Reaction



- Extent of reaction ξ : for $d\xi$, $dn_{\text{A}} = -d\xi$ and $dn_{\text{B}} = +d\xi$

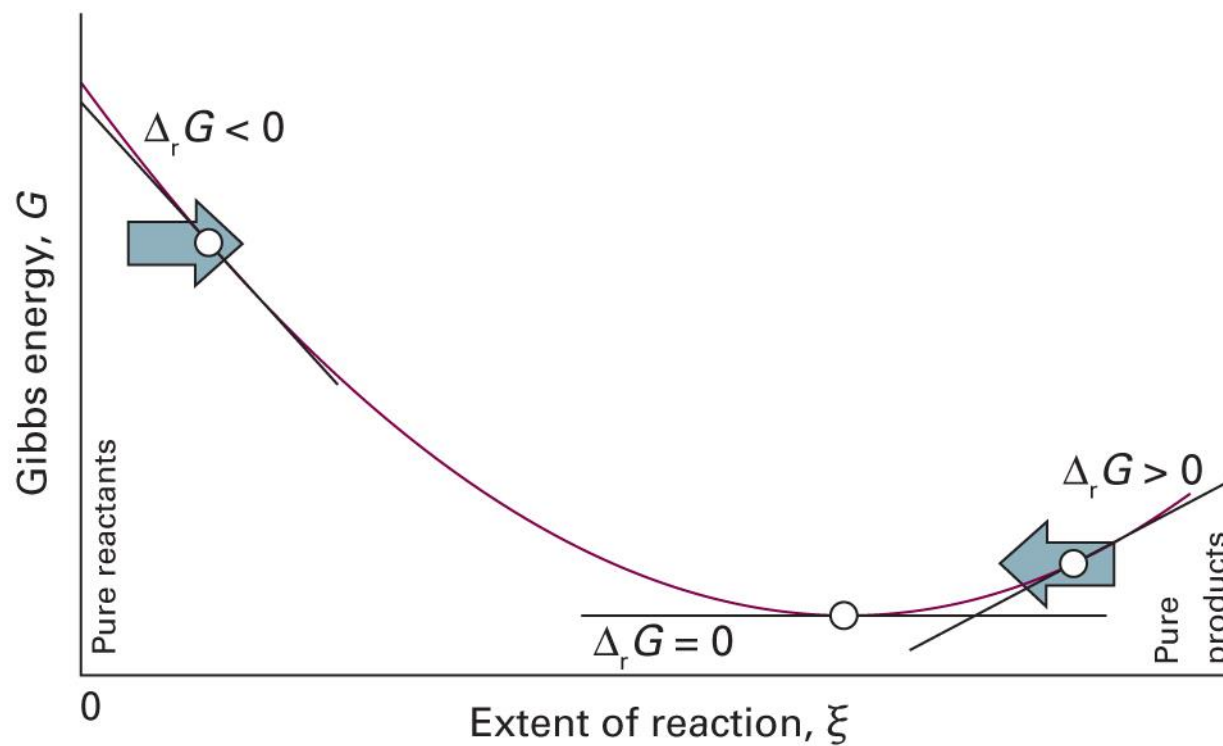
$$dG = \mu_{\text{A}}dn_{\text{A}} + \mu_{\text{B}}dn_{\text{B}} = -\mu_{\text{A}}d\xi + \mu_{\text{B}}d\xi = (\mu_{\text{B}} - \mu_{\text{A}})d\xi$$

- Reaction Gibbs energy

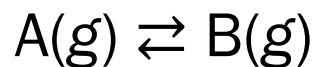
$$\Delta_{\text{r}}G = \left(\frac{\partial G}{\partial \xi} \right)_{p,T} = \mu_{\text{B}} - \mu_{\text{A}}$$

Progress of Reaction

$$\Delta_r G = \mu_B - \mu_A$$



Reaction Quotient



For a perfect gas, $\mu = \mu^\ominus + RT \ln(p/p^\ominus)$

$$\Delta_r G = \mu_B - \mu_A = \left\{ \mu_B^\ominus + RT \ln \left(\frac{p_B}{p_B^\ominus} \right) \right\} - \left\{ \mu_A^\ominus + RT \ln \left(\frac{p_A}{p_A^\ominus} \right) \right\}$$

Hence, using $\Delta_r G^\ominus = \mu_B^\ominus - \mu_A^\ominus$,

$$\Delta_r G = \Delta_r G^\ominus + RT \ln \left(\frac{p_B}{p_A} \right) = \Delta_r G^\ominus + RT \ln Q$$

Equilibrium Constant

$$\Delta_r G = \Delta_r G^\ominus + RT \ln Q$$

At equilibrium, $\Delta_r G = 0$

$$0 = \Delta_r G^\ominus + RT \ln Q_{\text{eq}} = \Delta_r G^\ominus + RT \ln K$$

Hence,

$$\Delta_r G^\ominus = -RT \ln K, \text{ where } K = (p_B/p_A)_{\text{eq}}$$

Generalization

For an arbitrary reaction,

$$\sum_{\text{reactant } i} \nu(i)i \rightleftharpoons \sum_{\text{product } j} \nu(j)j$$



$$\nu_A = 2, \nu_B = 1, \nu_C = 3, \nu_D = 1$$

Now,

$$dG = \sum_{\text{product } j} \mu_j dn_j + \sum_{\text{reactant } i} \mu_i dn_i$$

Generalization

$$dG = \sum_{\text{product } j} \mu_j dn_j + \sum_{\text{reactant } i} \mu_i dn_i$$

$$dG = \sum_{\text{product } j} \mu_j \nu_j d\xi - \sum_{\text{reactant } i} \mu_i \nu_i d\xi$$

$$dG = \left(\sum_{\text{product } j} \mu_j \nu_j - \sum_{\text{reactant } i} \mu_i \nu_i \right) d\xi$$

$$\Delta_r G = \left(\frac{\partial G}{\partial \xi} \right)_{p,T} = \sum_{\text{product } j} \mu_j \nu_j - \sum_{\text{reactant } i} \mu_i \nu_i$$

Chemical Potential and Activity

- Real gas: $\mu = \mu^\ominus + RT \ln(f/p^\ominus)$
- Real solution (solvent): $\mu = \mu^* + RT \ln a$
- Real solution (solute): $\mu = \mu^\ominus + RT \ln a$
- General: $\mu = \mu^\ominus + RT \ln a$

Component	Basis	Standard state	Activity	Limits
Solid or liquid		Pure, 1 bar	$a = 1$	
Solvent	Raoult	Pure solvent, 1 bar	$a = p/p^*, a = \gamma x$	$\gamma \rightarrow 1$ as $x \rightarrow 1$ (pure solvent)
Solute	Henry	(1) A hypothetical state of the pure solute	$a = p/K, a = \gamma x$	$\gamma \rightarrow 1$ as $x \rightarrow 0$
		(2) A hypothetical state of the solute at molality $b^\ominus$	$a = \gamma b/b^\ominus$	$\gamma \rightarrow 1$ as $b \rightarrow 0$
Gas	Fugacity [†]	Pure, a hypothetical state of 1 bar and behaving as a perfect gas	$f = \gamma p$	$\gamma \rightarrow 1$ as $p \rightarrow 0$



Generalization

$$\Delta_r G = \sum_{\text{product } j} \nu_j \mu_j - \sum_{\text{reactant } i} \nu_i \mu_i$$

For each species x ,

$$\nu_x \mu_x = \nu_x (\mu_x^\ominus + RT \ln a_x) = \nu_x \mu_x^\ominus + RT \ln(a_x^{\nu_x})$$

Then

$$\begin{aligned} \sum_x \nu_x \mu_x &= \sum_x \nu_x \mu_x^\ominus + RT \sum_x \ln(a_x^{\nu_x}) \\ &= \sum_x \nu_x \mu_x^\ominus + RT \ln \prod_x a_x^{\nu_x} \end{aligned}$$

Generalization

$$\Delta_r G = \sum_{\text{product } j} \nu_j \mu_j - \sum_{\text{reactant } i} \nu_i \mu_i$$

Substitution leads to

$$\Delta_r G = \Delta_r G^\ominus + RT \ln \prod_{\text{product } j} a_j^{\nu_j} - RT \ln \prod_{\text{reactant } i} a_i^{\nu_i}$$

where $\Delta_r G^\ominus = \sum_{\text{product } j} \nu_j \mu_j^\ominus - \sum_{\text{reactant } i} \nu_i \mu_i^\ominus$

$$\Delta_r G = \Delta_r G^\ominus + RT \ln \left(\prod_{\text{product } j} a_j^{\nu_j} / \prod_{\text{reactant } i} a_i^{\nu_i} \right)$$

Q and K

$$\Delta_r G = \Delta_r G^\ominus + RT \ln Q$$

where the **reaction quotient** Q is defined as

$$Q = \frac{\prod_{\text{product } j} a_j^{v_j}}{\prod_{\text{reactant } i} a_i^{v_i}}$$

At equilibrium, $\Delta_r G = 0$

$$0 = \Delta_r G^\ominus + RT \ln Q_{\text{eq}} = \Delta_r G^\ominus + RT \ln K$$

where the **equilibrium constant** K is Q at equilibrium

Brief Illustration 6A.3

Consider the reaction $2\text{A} + 3\text{B} \rightarrow \text{C} + 2\text{D}$. What is the reaction quotient?

Here, $\nu_{\text{A}} = 2$, $\nu_{\text{B}} = 3$, $\nu_{\text{C}} = 1$, and $\nu_{\text{D}} = 2$.

Using the definition of Q ,

$$Q = \frac{\prod_{\text{product } j} a_j^{\nu_j}}{\prod_{\text{reactant } i} a_i^{\nu_i}} = \frac{a_{\text{C}} a_{\text{D}}^2}{a_{\text{A}}^2 a_{\text{B}}^3}$$

Connection to Thermochemistry

$$\Delta_r G^\ominus = -RT \ln K$$

Since we know

$$\Delta_r G^\ominus = \sum_{\text{products } j} \nu_j \Delta_f G^\ominus(j) - \sum_{\text{reactants } i} \nu_i \Delta_f G^\ominus(i),$$

we can calculate K from thermochemical data.

Activity Coefficients

For the solution-phase reaction $A + B \rightleftharpoons C + D$,

$$K = \frac{a_C a_D}{a_A a_B} = \frac{(\gamma_C [C]/c^\ominus)(\gamma_D [D]/c^\ominus)}{(\gamma_A [A]/c^\ominus)(\gamma_B [B]/c^\ominus)}$$

Typically, we assume $\gamma_i \approx 1$ to approximate

$$K \approx \frac{([C]/c^\ominus)([D]/c^\ominus)}{([A]/c^\ominus)([B]/c^\ominus)}$$

Approximation for Activity

State	Measure	Approximation for a_j	Definition
Solute	molality	$b_j/b^\ominus$	$b^\ominus = 1 \text{ mol kg}^{-1}$
	molar concentration	$[J]/c^\ominus$	$c^\ominus = 1 \text{ mol dm}^{-3}$
Gas phase	partial pressure	$p_j/p^\ominus$	$p^\ominus = 1 \text{ bar}$
Pure solid, liquid		1 (exact)	

Pure Solids and Liquids

$$\Delta G_m = \mu_f - \mu_i = \int_{p_i}^{p_f} V_m dp$$

Let the initial state to be the standard state:

$$\mu - \mu^\ominus = \int_{p^\ominus}^p V_m dp' \approx V_m \int_{p^\ominus}^p 1 dp'$$

From the definition of an activity, $\mu - \mu^\ominus = RT \ln a$,

$$\ln a = \frac{V_m}{RT} \int_{p^\ominus}^p 1 dp' = \frac{V_m}{RT} (p - p^\ominus)$$

Not in Atkins

Example: Activity of Coke

The density of coke at 1000°C is about 1.5 g cm⁻³. Calculate the activity of C(s) in the form of coke at 100 bar and 1000°C.

The molar volume is

$$V_m = (\text{molar mass})/(\text{density}) = (12 \text{ g/mol})/(1.5 \text{ g cm}^{-3}) = 8.0 \text{ cm}^3 \text{ mol}^{-1}$$

Thus,

$$\begin{aligned}\ln a &= V_m(p - p^\ominus)/RT \\ &= (8.0 \text{ cm}^3 \text{ mol}^{-1}) \times (100 - 1) \text{ bar} / (0.082 \text{ L bar K}^{-1} \text{ mol}^{-1}) / (1273 \text{ K}) \\ &= 0.0076\end{aligned}$$

or $a = 1.01$.



Pure Solids and Liquids

$$\ln a = \frac{V_m}{RT} (p - p^\ominus)$$

At moderate pressures, the activity is very close to 1.

Hence, we usually take $a = 1$ for pure solids and liquids.

For Gaseous Reactions

$$K = \frac{\prod_{\text{product } j} a_j^{\nu_j}}{\prod_{\text{reactant } i} a_i^{\nu_i}} = \frac{\prod_{\text{product } j} (p_j/p^\ominus)^{\nu_j}}{\prod_{\text{reactant } i} (p_i/p^\ominus)^{\nu_i}}$$

Since $p_i/p^\ominus = [i](RT/p^\ominus) = [i]/c^\ominus (c^\ominus RT/p^\ominus)$,

$$K = \frac{\prod_{\text{product } j} ([j]/c^\ominus)^{\nu_j}}{\prod_{\text{reactant } i} ([i]/c^\ominus)^{\nu_i}} \left(\frac{c^\ominus RT}{p^\ominus} \right)^{\sum_j \nu_j - \sum_i \nu_i}$$

$$K = K_c \left(\frac{c^\ominus RT}{p^\ominus} \right)^{\Delta \nu} = K_c \left(\frac{T}{12.03 \text{ K}} \right)^{\Delta \nu}$$

Brief Illustration 6A.5

What is the relationship between K and K_c for the reaction $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightarrow 2\text{NH}_3(\text{g})$ at 298.15 K?

$$\Delta\nu = 2 - 3 - 1 = -2$$

Hence,

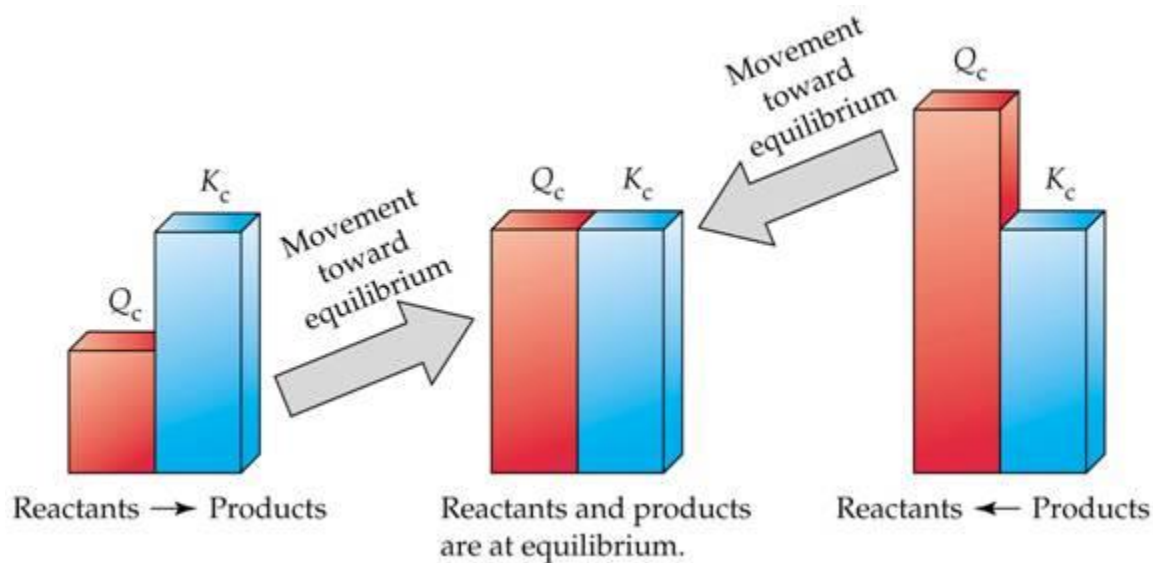
$$K = K_c \left(\frac{T}{12.03 \text{ K}} \right)^{\Delta\nu} = K_c \left(\frac{298.15 \text{ K}}{12.03 \text{ K}} \right)^{-2} = \frac{K_c}{614.2}$$

Meaning of Q and K

- Reaction quotient Q
 - “How far the equilibrium is”
 - Varies as the reaction progresses
 - $Q = K$ at equilibrium
- Equilibrium constant K
 - Amounts of reactants and products at equilibrium
 - Constant for the given experimental condition

Not in Atkins

Meaning of Q and K



Le Chatelier's Principle

When subjected to a disturbance,
a system at equilibrium tends to respond in a way
that **minimizes the effect of the disturbance**.

Disturbances to consider:

- Amount (concentration, pressure)
- Temperature

Amount Change

- **Increase** of the amount of a particular species
 - The system responds to **decrease** the amount
- **Decrease** of the amount of a particular species
 - The system responds to **increase** the amount
- Reactant increase / product decrease: $Q < K$
 - Decrease of reactants & increase of products
- Reactant decrease / product increase: $Q > K$
 - Increase of reactants & decrease of products

Volume Change

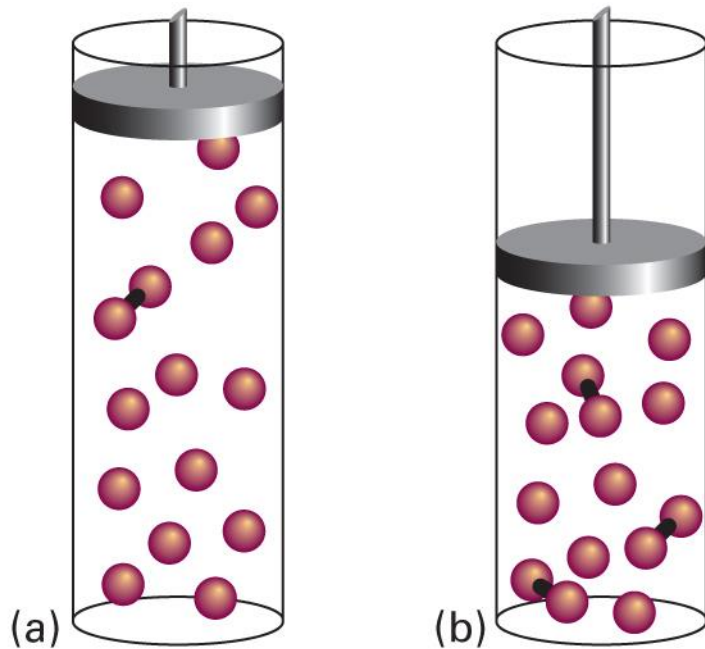
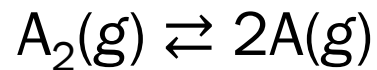
For a gaseous reaction,

- Increase of the volume → decrease of the total amount
- Decrease of the volume → increase of the total amount

(Think of concentrations)

Pressure change is linked to volume change via $pV = nRT$

Volume Change



Volume Change

$$K_c = \frac{\prod_{\text{product } j} ([j]/c^\ominus)^{\nu_j}}{\prod_{\text{reactant } i} ([i]/c^\ominus)^{\nu_i}} = \frac{\prod_{\text{product } j} (n_j/Vc^\ominus)^{\nu_j}}{\prod_{\text{reactant } i} (n_i/Vc^\ominus)^{\nu_i}}$$

For a volume change from V to xV ,

$$Q_c = \frac{\prod_{\text{product } j} (n_j/xVc^\ominus)^{\nu_j}}{\prod_{\text{reactant } i} (n_i/xVc^\ominus)^{\nu_i}} = K_c x^{-(\sum_j \nu_j - \sum_i \nu_i)} = K_c x^{-\Delta \nu}$$

Volume Change

$$Q_c = K_c x^{-\Delta\nu}$$

For $x > 1$ (volume increase),

- $\Delta\nu = 0$: $Q_c = K_c$ and the system is at equilibrium
- $\Delta\nu > 0$: $Q_c < K_c$ and the forward reaction is favored
- $\Delta\nu < 0$: $Q_c > K_c$ and the reverse reaction is favored

Temperature Change

- Exothermic reactions
 - Increased temperature favors the reactants
 - Decreased temperature favors the products
- Endothermic reactions
 - Increased temperature favors the products
 - Decreased temperature favors the reactants

van't Hoff Equation

$$\ln K = -\Delta_r G^\ominus / RT$$

Differentiating both sides with respect to T ,

$$\frac{d \ln K}{dT} = -\frac{1}{R} \frac{d(\Delta_r G^\ominus / T)}{dT}$$

Using the Gibbs-Helmholtz equation,

$$-R \frac{d \ln K}{dT} = \frac{d(\Delta_r G^\ominus / T)}{dT} = -\frac{\Delta_r H^\ominus}{T^2}$$

$$\frac{d \ln K}{dT} = \frac{\Delta_r H^\ominus}{RT^2}$$

T*-Dependence of *K

$$\frac{d \ln K}{dT} = \frac{\Delta_r H^\ominus}{RT^2}$$

Integration gives

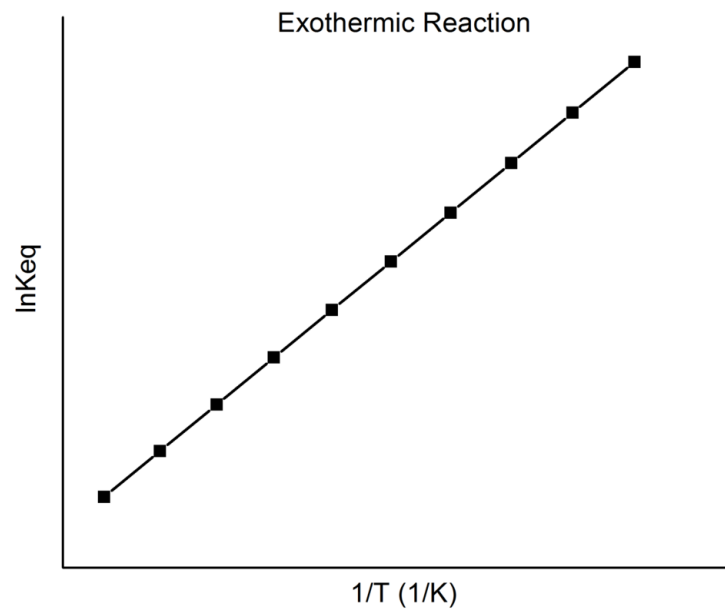
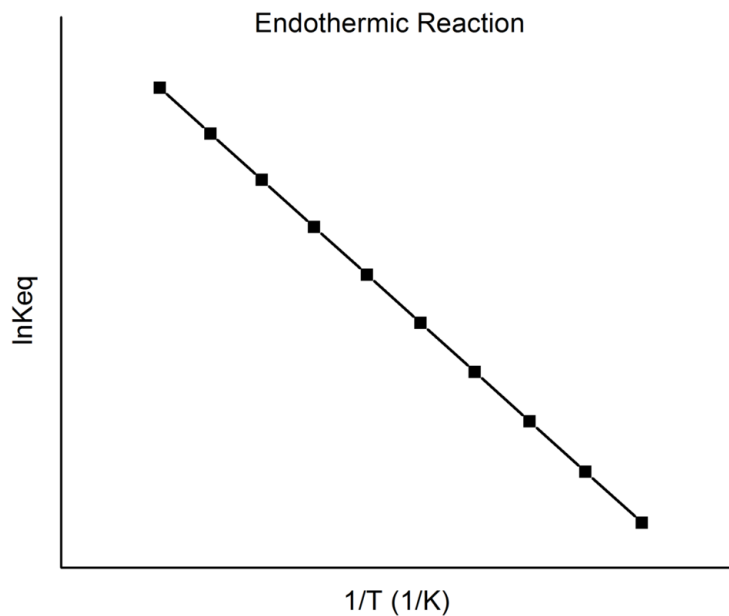
$$\int_{\ln K_1}^{\ln K_2} d \ln K = \int_{T_1}^{T_2} \frac{\Delta_r H^\ominus}{RT^2} dT$$

Assuming that $\Delta_r H^\ominus$ is constant,

$$\ln K_2 - \ln K_1 = \frac{\Delta_r H^\ominus}{R} \int_{T_1}^{T_2} \frac{1}{T^2} dT = -\frac{\Delta_r H^\ominus}{R} \left(\frac{1}{T_2} - \frac{1}{T_1} \right)$$

T-Dependence of *K*

$$\ln K_2 - \ln K_1 = -\frac{\Delta_r H^\ominus}{R} \left(\frac{1}{T_2} - \frac{1}{T_1} \right)$$



Summary

